Q.1]

```
import java.util.*;
Class Demo {
  Public static void main (String args[]) {
   Scanner scan = new Scanner (System.in);
    int n = Scan.nextInt();
      for (int i=1 ; i<=n ; i++) {
        for (int j=1 ; j<=n ; j++) {
          if (i==1 || i==n || j==1 || j==n)
          {
            sop ("* ");
          } else {
            sop (" _ ");
          }
        }  sopln();
      }
    }
```

output

```
   1| 2| 3| 4| 5
   *  *  *  *  *
2| *  _  _  _  *
3| *  _  _  _  *
4| *  _  _  _  *
5| *  *  *  *  *
```

if (j=1 || j=n || i=1 || j=n

i=1 i≤5	i=2 2<5	j=3 3<4
j=1 = True ★	j=1 = True - ★	j=1 - True
j=2 = ~~False~~ True ★	j=2 - False -	j=2 False -
j=3 = ~~False~~ True ★	j=3 - False -	j=3 False -
j=4 = ~~False~~ True ★	j=4 False -	j=4 False -
j=5 = ~~True~~ True ★	j=5 True ★	j=5 True -

i=4	i=n 5
j=1 True ★	j=1 = True ★
j=2 False -	j=2 = False -
j=3 False -	j=3 = False -
j=4 False -	j=4 = False -
j=5 True ★	j=5 = True ★

Q.2]

```
import java.util.scanner;
~~public~~ Class Demo {
 Scanner
 sop
 int n
  for( int i =1 ; i<=n ; i++){
      for( int j =1 ; j<=n ; j++){
          if (i==j)|| i+j ==(n+1)){
            sop (" *");
          }else {
            sop (" ");
          }
        }
        sopln();
      }
    }
```

output

```
*     *
 *   *
  *
 *   *
*     *
```

Page No.
Date: / /

Q.1]

Dry run $n=5$

$i=1$ $\boxed{1}$ $j=2$

$1 \le 5 \rightarrow$ True * $2 \le 5$ true

$j=1$ $\boxed{1}$ $i=1\ j=1 \rightarrow$ false

$i=1$ $j \le 5 \rightarrow$ true * $i=2=j=2 \rightarrow$ true $\boxed{i=j}$

$i=1$ $j=2$ × false $i=2\ j=3 \rightarrow$

$i=1$ $j=3$ × false $i=2^{+}\ j=4 \rightarrow$ true $= \boxed{n+1}$

$i=1$ $j=4$ × false $i=2\ j=5 \rightarrow$ false

$i=1$ $j=5 \rightarrow$ True *

$i=3$

$i=3$ $j=1 \rightarrow$ false

$i=3$ $j=2$ = =||=

$i=3$ $j=3$ = True $\boxed{i=j}$

$i=3$ $j=4$ = false

$i=3$ $j=5$ = ||=

$i=4$

$i=4\ j=1$ false

$i=4\ j=2$ true $\boxed{n+1}$

$i=4\ j=3$

$i=4\ j=4$ true $\boxed{i=j}$

$i=4\ j=5$

$i=5$

$i=5$ $j=1$ = true $\boxed{n+1}$

$i=5$ $j=2$

$i=5$ $j=3$ } false

$i=5$ $j=4$

$i=5$ $j=5$ = true $\boxed{i=j}$

$i=6$

$6 \le 5$

False

Q.3]

```
import java.util.*;
class Demo {
Public static void main (String args[]){
scanner scan = new Scanner (System.in);
sop ("Enter a value of n");
int n = scan.nextInt();
for (int i = 1 ; i ≤ n ; i++) {
    for (int j = 1 ; j ≤ 5 ; j++) {
        if (i==j || i>j) {
        sop ("*");
    } else {
        sop (" ");
    }
    }
    sopln();
}
}
}
```

output

```
*
* *
* * *
* * * *
* * * * *
```

if (i==j) || i>j

i≤n = ~~n~~ n=5

int i = 1

1 ≤ 5

~~int i=2~~ i=2 ~~int i=3~~

~~int~~ i=1 j=1 = true ~~i=2~~

i=1 j=2 = true

i=1 j=3 = $\boxed{\text{true}}$ -> ~~true~~

i=1 j=4 = true

i=1 j=5 = true

~~int~~ i=2 j=1 false

i=2 j=2 true

i=2 j=3 true

i=2 j=4 true

i=2 j=5 true

i=3

i=3 j=1 = false

i=3 j=2 = false

i=3 j=3 = true

i=3 j=4 = true

i=3 j=5 = true

i=4

i=4 j=1 = false

i=4 j=2 = false

i=4 j=3 = false

i=4 j=4 = true

i=4 j=5 = true

i=5

~~int~~ 5 j=1 = false

~~int~~ 5 j=1 = false

~~int~~ 5 j=1 = false

~~int~~ 5 j=1 = false

i=5 j=1 = true

$\boxed{i=6}$ $\boxed{6 \leq 5}$

false

Q.4]

```
import java.util.*;
class Demo {
 public static void main (String args[])
 Scanner Scan = new Scanner (System.in)
  sop(" Enter a value of n ");
   int n = scan.nextInt();
For (int i=1 ; i<=n ; i++) {
      for ( int j =1 ; j<5 ; j++) {
          if (i+j <= n+1) {
              sop (" * ");
          } else {
              sop(" ");
          }
      }
      sopln();
    }
  }
}
```

```
* * * * *
* * * *
* * *
* *
*
```

if $(i+j <= n+1)$

$n = 5$

int $i = 1$ $i \leq 5$

$i = 1$ $j = 1 \rightarrow$ true
$i = 1$ $j = 2 \rightarrow$ true
$i = 1$ $j = 3 \Rightarrow$ true
$i = 1$ $j = 4 \Rightarrow$ true
$i = 1$ $j = 5 \Rightarrow$ true

$i = 2$

$i = 2$ $j = 1 \Rightarrow$ true
$i = 2$ $j = 2 \Rightarrow$ true
$i = 2$ $j = 3 \Rightarrow$ true
$i = 2$ $j = 4 \Rightarrow$ true
$i = 2$ $j = 5 \Rightarrow$ false

$i = 3$

$i = 3$ $j = 1 \Rightarrow$ true
$i = 3$ $j = 2 \Rightarrow$ true
$i = 3$ $j = 3 \Rightarrow$ true
$i = 3$ $j = 4 \Rightarrow$ ~~true~~ false
$i = 3$ $j = 5 \Rightarrow$ false

$i = 4$

$i = 4$ $j = 1 \Rightarrow$ true
$i = 4$ $j = 2 \Rightarrow$ true
$i = 4$ $j = 3 \Rightarrow$ false
$i = 4$ $j = 4 \Rightarrow$ false
$i = 4$ $j = 5 \Rightarrow$ false

$i = 6$ $\boxed{6 \leq 5}$

$i = 5$

$i = 5$ $j = 1 \Rightarrow$ true
$i = 5$ $j = 2 \Rightarrow$ false
$i = 5$ $j = 3 \Rightarrow$ false
$i = 5$ $j = 4 \Rightarrow$ false
$i = 5$ $j = 5 \Rightarrow$ false

5]

```
import java.util.*;
class Demo{
public static void main(String args[]){
Scanner scan = new Scanner(System.in);
~~int n~~ sop("Enter a value of:");
int n = scan.nextInt();
  for (int i = 1; i<=n; i++){
    for (int j = 1; j<=n; j++){
    if(((i+j) % 2==0 || n % 2==0) &&
      ((i+j) % 2 != 0 || n % 2 != 0)){
      sop("#");
    } else {
      sop("*");
    }
  }
```

output

```
* # * # *
# * # * #
* # * # *
# * # * #
* # * # *
```

if (i+j%2==0

n=5

int i = 1

i≤5

i=1 j=1 = False

i=1 j=2 ⇒ true

i=1 j=3 ⇒ False

i=1 j=4 ⇒ true

i=1 j=5 ⇒ False

i=3

3≤5

i=3 j=1 = False

i=3 j=2 = true

i=3 j=3 = False

i=3 j=4 = true

i=3 j=5 = False

i=5

5≤5

i=5 j=1 = False

i=5 j=2 = true

i=5 j=3 = False

i=5 j=4 = true

i=5 j=5 = False

i=2

2≤5

i=2 j=1 = true

i=2 j=2 = False

i=2 j=3 = true

i=2 j=4 = False

i=2 j=5 = true

i=4

4≤5

i=4 j=1 = true

i=4 j=2 = False

i=4 j=3 = true

i=4 j=4 = False

i=4 j=5 = true